

LECTURE 18: SYNTHETIC CONTROL METHODS: BASICS

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INTRODUCTION

- California's anti-smoking legislation (Proposition 99) took effect in 1989.
- What is the causal effect of the legislation on smoking rates in California in 1989?
- We **observe** smoking rates in California in 1989 given the legislation. We need to **impute** the **counterfactual** smoking rates in California in 1989 had the legislation not been enacted.
- We have data in the absence of smoking legislation in California prior to 1989, and for other states both before and in 1989. (And possibly other variables, but not of essence)

ANOTHER EXAMPLE

- California changed the tax treatment of out-of-state online sales.
- How did that change online sales in California?
- Sales data on other states before and after, and California before, with strong trends in sales.

GENERAL SET UP

$$\mathbf{Y} = \begin{pmatrix} Y_{11} & Y_{12} & Y_{13} & \dots & Y_{1T} \\ Y_{21} & Y_{22} & Y_{23} & \dots & Y_{2T} \\ Y_{31} & Y_{32} & Y_{33} & \dots & Y_{3T} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Y_{N1} & Y_{N2} & Y_{N3} & \dots & Y_{NT} \end{pmatrix} \quad \text{(realized outcome).}$$

$$\mathbf{W} = \begin{pmatrix} 1 & 1 & 0 & \dots & 1 \\ 0 & 0 & 1 & \dots & 0 \\ 1 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 1 & \dots & 0 \end{pmatrix} \quad \text{(binary treatment).}$$

- Rows of $\mathbf{Y}$ and $\mathbf{W}$ correspond to units (e.g., states),
- Columns correspond to time periods (e.g., years).

IN TERMS OF POTENTIAL OUTCOME MATRICES $\mathbf{Y}(C)$ AND $\mathbf{Y}(T)$

$$\mathbf{Y}(C) = \begin{pmatrix} ? & ? & \checkmark & \dots & ? \\ \checkmark & \checkmark & ? & \dots & \checkmark \\ ? & \checkmark & ? & \dots & \checkmark \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ ? & \checkmark & ? & \dots & \checkmark \end{pmatrix}$$

$$\mathbf{Y}(T) = \begin{pmatrix} \checkmark & \checkmark & ? & \dots & \checkmark \\ ? & ? & \checkmark & \dots & ? \\ \checkmark & ? & \checkmark & \dots & ? \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \checkmark & ? & \checkmark & \dots & ? \end{pmatrix}$$

- $Y_{it} = (1 - W_{it})Y_{it}(0) + W_{it}Y_{it}$.
- We are interested in the average treatment effect for the treated, (or other average, e.g., overall average effect)

$$\tau = \frac{\sum_{i,t} W_{it} (Y_{it}(T) - Y_{it}(C))}{\sum_{i,t} W_{it}},$$

- We need to **impute** the missing potential outcomes in $\mathbf{Y}(C)$ (and in $\mathbf{Y}(T)$ for other estimands).

STRUCTURE ON ASSIGNMENT PROCESS

- Often structure on **W**: treated unit/time-period block

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 1 \\ 0 & 0 & 0 & \dots & 1 & 1 \end{pmatrix}$$

- Staggered adoption (e.g., adoption of technology, Athey and Stern, 1998)

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots & 0 & \text{(never adopter)} \\ 0 & 0 & 0 & 0 & \dots & 1 & \text{(late adopter)} \\ 0 & 0 & 1 & 1 & \dots & 1 & \\ 0 & 0 & 1 & 1 & \dots & 1 & \text{(medium adopter)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \\ 0 & 1 & 1 & 1 & \dots & 1 & \text{(early adopter)} \end{pmatrix}$$

EXAMPLE I: THE MARIEL BOATLIFT STUDY

- One of the classic applications of difference-in-differences methods is the Mariel Boatlift study by Card (1990). Card studies the effect of the influx of low-skilled labor into the Miami labor market on wages using data on other labor markets for comparison.
- Recently this study has been revisited using synthetic control methods in Peri and Yasenov (2015).
- The Peri and Yasenov (2015) study uses
 - $N_T = 1$ treated unit,
 - $N_C = 44$ potential control units,
 - $T_0 = 7$ pre-treatment periods
 - $T_1 = 6$ post-treatment periods.

EXAMPLE II: THE CARD-KRUEGER NEW-JERSEY PENNSYLVANIA MINIMUM WAGE STUDY

- In the Card and Krueger (1994) study, the focus is on the causal effect of a change in the minimum wage in New Jersey. Card and Krueger use data from fast food restaurants in New Jersey and Pennsylvania.
- They use information on
 - $N_T = 321$ treated (new Jersey) units,
 - $N_C = 78$ control (Pennsylvania) units,
 - one pre-treatment period, $T_0 = 1$,
 - one post treatment period, $T_1 = 1$.

EXAMPLE III: THE CALIFORNIA SMOKING LEGISLATION STUDY

- In the key study on synthetic control methods, Abadie, Diamond and Hainmueller (2010) focus on estimating the effect of anti-smoking legislation in California. It uses smoking per capita as the outcome.
- Abadie, Diamond and Hainmueller (2010) use information on
 - $N_T = 1$ treated unit,
 - $N_C = 29$ control states without such anti-smoking measures,
 - $T_0 = 17$ pre-program years
 - $T_1 = 13$ post-program years.

EXAMPLE IV: THE GERMAN RE-UNIFICATION STUDY

- In another classic synthetic control application Abadie, Diamond and Hainmueller (2014) study the effect on per capita Gross Domestic Product in West-Germany of the re-unification with East Germany.
- They use
 - $N_T = 1$ treated unit,
 - $N_C = 16$ countries as potential controls
 - $T_0 = 30$ years of data prior to re-unification
 - $T_1 = 14$ years of data post re-unification.

THE SINGLE TREATED UNIT/TIME-PERIOD CASE

- We focus for the moment on case with a single treated unit/time-period

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{pmatrix}$$

- **Challenge:** Trying to predict $Y_{NT}(C)$ based on observed values $Y_{it}(C)$ for $(i, t) \neq (N, T)$ (not using $Y_{NT}(T)$).

THREE RELATED LITERATURES

- synthetic control literature (vertical regression)
- causal literature with unconfoundedness (horizontal regression)
- difference-in-differences, two-way-fixed-effect methods, and factor models
- see Arkhangelsky & Imbens survey for framing of the literature.
- See Abadie (2021) for survey of SC literature.

ABADIE-DIAMOND-HAINMUELLER SYNTHETIC CONTROL METHOD

- ADH suggest using a weighted average of outcomes for other states:

$$\hat{Y}_{NT}(0) = \sum_{j=1}^{N-1} \omega_j Y_{jT}$$

- ADH restrict the weights ω_j to simplex:
 - weights are non-negative,
 - weights sum to one.
- Weight restrictions play two roles:
 - substantive: restrictions make sense substantively
 - regularization: helps estimation when there are a lot of control units.
- Let $Y_{i,L}$ be the vector of lagged values $Y_{i,t}$ for $t < T$.

ABADIE-DIAMOND-HAINMUELLER SYNTHETIC CONTROL METHOD

- Let Z_i be the vector of functions of covariates X_{it} , including possibly some lagged outcomes Y_{it} for pre- T periods (but not necessarily all pre-treatment outcomes).
- Let the norm be $\|a\|_V = a'V^{-1}a$
- ADH first solve

$$\omega(V) = \arg \min_{\omega} \left\| Z_N - \sum_{i=1}^{N-1} \omega_i \cdot Z_i \right\|_V$$

- This finds, for a given weight matrix V the optimal weights.
- But: **how do we choose weight matrix V ?**
- Equal weights **does not make sense**, would not be invariant to linear transformations.

ABADIE-DIAMOND-HAINMUELLER SYNTHETIC CONTROL METHOD

- ADH find positive semi-definite V that minimizes distance between pre-treatment outcome vector for treated unit and weighted average for control units:

$$\hat{V} = \arg \min_V \left\| Y_{N,L} - \sum_{i=1}^{N-1} \omega_i(V) \cdot Y_{i,L} \right\|$$

- Then the weights ω are based on using weight matrix $\hat{V}$:

$$\omega = \omega(\hat{V}) = \arg \min_{\omega} \left\| Z_N - \sum_{i=1}^{N-1} \omega_i \cdot Z_i \right\|_{\hat{V}}$$

ABADIE-DIAMOND-HAINMUELLER SYNTHETIC CONTROL METHOD

- Suppose $Z_i = Y_{i,L}$ (only the lagged outcomes), then:

$$\omega = \arg \min_{\omega} \left\| Y_{N,L} - \sum_{i=1}^{N-1} \omega_i \cdot Y_{i,L} \right\| = \arg \min_{\omega} \sum_{t=1}^{T-1} \left(Y_{N,t} - \sum_{i=1}^{N-1} \omega_i \cdot Y_{i,t} \right)^2$$

subject to simplex restrictions

$$\omega_i \geq 0, \quad \sum_{i=1}^{N-1} \omega_i = 1$$

DOUDCHENKO-IMBENS

- Alternative Characterization as Vertical Regression
- Abadie-Diamond-Hainmueller estimator:

$$\hat{\tau}_{ADH} = Y_{NT} - \hat{Y}_{NT}, \quad \hat{Y}_{NT} = \sum_{i=1}^{N-1} \omega_i Y_{iT}$$

- Solution for ADH weights can be characterized in single regression step in absence of covariates:

$$\omega = \arg \min_{\omega} \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it} \right)^2,$$

subject to simplex restriction

$$\omega_i \geq 0, \quad \sum_{i=1}^{N-1} \omega_i = 1.$$

DOUDCHENKO-IMBENS

- Modification: Doudchenko-Imbens:

$$\hat{\tau}_{DI} = Y_{NT} - \hat{Y}_{NT}, \quad \hat{Y}_{NT} = \alpha + \sum_{i=1}^{N-1} \omega_i Y_{iT}$$

- Now: regression with intercept

$$(\alpha, \omega) = \arg \min_{\alpha, \omega} \sum_{t=1}^{T-1} \left(Y_{Nt} - \alpha - \sum_{i=1}^{N-1} \omega_i Y_{it} \right)^2 + \text{regularization term},$$

with no restrictions on ω .

- ADH cannot accommodate intercept and need adding up restriction because they also match weighted sum of covariates in control units to treated unit.
- Regularization is important here if N is large relative to T , partly because of lack of restrictions on ω .

COMPARISON TO UNCONFOUNDEDNESS METHODS / HORIZONTAL REGRESSION

- Alternative based on unconfoundedness type assumption:

$$\hat{\tau}_{\text{UNC}} = Y_{NT} - \hat{Y}_{NT}, \quad \hat{Y}_{NT} = \alpha + \sum_{t=1}^{T-1} \lambda_t Y_{Nt}$$

- Here

$$(\alpha, \lambda) = \arg \min_{\alpha, \lambda} \sum_{i=1}^{N-1} \left(Y_{iT} - \alpha - \sum_{t=1}^{T-1} \lambda_t Y_{it} \right)^2 + \text{regularization term},$$

- No restrictions on λ .
- (One could impose restrictions on λ and rule out intercept, but this is not done in practice.)

UNCONFOUNDEDNESS METHODS / HORIZONTAL REGRESSION (SLIGHT MODIFICATION)

- Slightly unusual unconfoundedness regression:

$$\hat{\tau}_{\text{UNC}} = Y_{NT} - \hat{Y}_{NT}, \quad \hat{Y}_{NT} = \sum_{t=1}^{T-1} \beta_t Y_{Nt}$$

$$\beta = \lim_{\lambda \rightarrow 0} \arg \min_{\beta} \sum_{i=1}^{N-1} \left(Y_{iT} - \sum_{t=1}^{T-1} \beta_t Y_{it} \right)^2 + \lambda \|\beta\|_2$$

- Similar estimator for SC:

$$\hat{\tau}_{\text{ADH}} = Y_{NT} - \hat{Y}_{NT}, \quad \hat{Y}_{NT} = \sum_{i=1}^{N-1} \omega_i Y_{iT}$$

$$\omega = \lim_{\lambda \rightarrow 0} \arg \min_{\omega} \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} \omega_t Y_{it} \right)^2 + \lambda \|\omega\|_2$$

RELATION BETWEEN HORIZONTAL AND VERTICAL REGRESSION

- Shen, Ding, Sekhon and Yu (2022):

$$\hat{\tau}_{\text{ADH}} = \hat{\tau}_{\text{UNC}}$$

- Does **not** hold if we add an intercept to unconfoundedness regression.
- Does **not** hold if we impose simplex (nonnegativity and additivity) restrictions on weights.

DIFFERENCE-IN-DIFFERENCES

- Estimate causal effect as $\hat{\tau}^{\text{DID}}$, defined by

$$\left(\hat{\tau}^{\text{DID}}, \hat{\alpha}, \hat{\gamma}\right) = \arg \min_{\alpha, \gamma, \tau} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})^2$$

FACTOR MODELS

- More general, **factor models**
- . DID/TWFE corresponds to specification

$$Y_{it}(C) = \alpha_i + \beta_t + \varepsilon_{it}$$

- We could generalize this to a factor model with R factors:

$$Y_{it}(C) = \sum_{r=1}^R \gamma_{tr} \delta_{ir} + \varepsilon_{it} \quad \mathbf{Y}(0) = \Gamma \Delta^\top + \mathbf{E}$$

- To see this is a strict generalization of the two-way-fixed-effect model if $R \geq 2$, suppose $\gamma_{t1} = 1, \gamma_{t2} = \beta_t, \delta_{i1} = \alpha_i, \delta_{i2} = 1$, then:

$$Y_{it}(C) = \sum_{r=1}^R \gamma_{tr} \delta_{ir} + \varepsilon_{it} = \gamma_{t1} \delta_{i1} + \gamma_{t2} \delta_{i2} + \varepsilon_{it} = \alpha_i + \beta_t + \varepsilon_{it}$$

- But clearly **more flexible than TWFE** in general, especially if $R > 2$.

SINGULAR VALUE DECOMPOSITION

- Singular value decomposition:

$$\mathbf{Y}_{N \times T} = \mathbf{U}_{N \times N} \Sigma_{N \times T} \mathbf{V}_{T \times T}^T,$$

- Here $\mathbf{U}$ and $\mathbf{V}$ are orthonormal matrices, and Σ has non-zero elements (the **singular values**) only on the main diagonal.
- If $N \leq T$, this means we can fit the matrix $\mathbf{Y}(C)$ **perfectly** with N factors, and $\gamma_{tr} = \sum_r r r V_{tr}$ and $\delta_{ir} = U_{ir}$.
- So, we need to **limit** the rank of the approximation. We can fix the rank *ex ante*, or **regularize** it by minimizing

$$\sum_{i,t} \left(y_{it} - \sum_{r=1}^R \gamma_{tr} \delta_{ir} \right)^2 + \lambda \|\Gamma \Delta^T\|_{\text{nn}},$$

- Here the **nuclear norm** of $\Gamma \Delta'$ is the sum of the singular values.
- We can choose the tuning parameter λ by cross-validation.

INCLUDING UNIT AND TIME FIXED EFFECTS

- With regularization on factors it may be helpful to still include unit and time fixed effects without regularization:

$$\left(\hat{\tau}^{\text{FM}}, \hat{\alpha}, \hat{\gamma}\right) = \arg \min_{\alpha, \gamma, \tau} \sum_{i=1}^N \sum_{t=1}^T \left(Y_{it} - \alpha_i - \beta_t - \sum_{r=1}^R \gamma_{tr} \delta_{ir} - \tau W_{it} \right)^2$$

+regularization term(γ, δ)

(Athey, Bayati, Doudchenko, Imbens, Khosravi, 2018)

CHOOSING BETWEEN THESE METHODS

- Challenge: How to choose between these methods (vertical/horizontal regression, factor models), or how to tie them together?
- Observation I synthetic control can be characterized as

$$\hat{\tau}^{\text{ADH}} = \arg \min_{\tau, \gamma} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \tau W_{it}) \times \omega_i^{\text{ADH}}$$

- Regression with time fixed effects and ADH weights (easy to include covariates).
- For this representation it is important to have weights restricted – Doudchenko-Imbens estimator cannot be written in this regression form.
- Under some conditions standard errors can be based on regression interpretation taking weights as given (even though the weights depend on outcome data).

CHOOSING BETWEEN THESE METHODS

- **Observation II** did/twfe can be characterized as

$$\hat{\tau}^{\text{DID}} = \arg \min_{\tau, \gamma, \alpha} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})$$

- regression with time fixed effects and unit fixed effects, no weights.

SYNTHETIC DIFFERENCE IN DIFFERENCES (SDID)

- Generalization:

$$\hat{\tau}^{\text{SDID}} = \arg \min_{\tau, \gamma, \alpha} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it}) \times \omega_i^{\text{ADH}} \times \lambda_t^{\text{ADH}}$$

- Regression with unit and time fixed effects, and with unit and time weights.
- Time weights satisfy:

$$\lambda = \arg \min_{\lambda} \sum_{i=1}^{N-1} \left(Y_{iT} - \sum_{t=1}^{T-1} \lambda_t Y_{it} \right)^2 + \text{regularization term},$$

- with simplex restriction

$$\lambda_t \geq 0, \quad \sum_{t=1}^{T-1} \lambda_t = 1.$$

(or down-weight observations from distant past.)

FURTHER GENERALIZATION

- Generalization: factor models unit and time fixed effects with synthetic control unit and time weights.

$$\hat{\tau}^{\text{FM}} = \arg \min_{\alpha, \gamma, \tau} \sum_{i=1}^N \sum_{t=1}^T \left(Y_{it} - \delta_i - \beta_t - \sum_{r=1}^R \gamma_{tr} \alpha_{ir} - \tau W_{it} \right)^2 \omega_i^{\text{ADH}} \lambda_t^{\text{ADH}}$$

DOUBLE BIAS REDUCTION

- DID is double bias reduction method:

$$\begin{aligned}\hat{Y}_{NT}^{\text{DID}}(0) &= \frac{1}{(N-1)(T-1)} \sum_{i=1}^{N-1} \sum_{t=1}^{T-1} Y_{it} \\ &+ \frac{1}{N-1} \sum_{i=1}^{N-1} \left(Y_{iT} - \frac{1}{T-1} \sum_{t=1}^{T-1} Y_{it} \right) \quad \text{time bias} \\ &+ \frac{1}{T-1} \sum_{t=1}^{T-1} \left(Y_{Nt} - \frac{1}{N-1} \sum_{i=1}^{N-1} Y_{it} \right) \quad \text{unit bias}\end{aligned}$$

INTERPRETATION OF SDID

- ADH is single bias reduction method, but with local weights:

$$\hat{Y}_{NT}^{\text{ADH}}(0) = \frac{1}{T-1} \sum_{i=1}^{N-1} \sum_{t=1}^{T-1} \omega_i^{\text{ADH}} Y_{it}$$
$$+ \sum_{i=1}^{N-1} \omega_i^{\text{ADH}} \left(Y_{iT} - \frac{1}{T-1} \sum_{t=1}^{T-1} Y_{it} \right) \quad \text{time bias.}$$

INTERPRETATION OF SDID

- SDID is double bias reduction method, with local weights for units and time:

$$\begin{aligned}\hat{Y}_{NT}^{SDID}(0) &= \frac{1}{(N-1)(T-1)} \sum_{i=1}^{N-1} \sum_{t=1}^{T-1} Y_{it} \omega_i^{ADH} \lambda_t^{ADH} \\ &+ \frac{1}{N-1} \sum_{i=1}^{N-1} \omega_i^{ADH} \left(Y_{iT} - \frac{1}{T-1} \sum_{t=1}^{T-1} \lambda_t^{ADH} Y_{it} \right) \quad \text{time bias} \\ &+ \frac{1}{T-1} \sum_{t=1}^{T-1} \lambda_t^{ADH} \left(Y_{Nt} - \frac{1}{N-1} \sum_{i=1}^{N-1} \omega_i^{ADH} Y_{it} \right) \quad \text{unit bias}\end{aligned}$$

- Double Robustness
 - If a factor model holds, but the weights are good (e.g., ADH weights), SDID is consistent.
 - If the DID model holds, but we use arbitrary weights, SDID is consistent.

CALIFORNIA SMOKING DATA CALCULATIONS

- Take pre-1988 data for all states, so we observe all $Y_{it}(0)$ for all unit/time pairs.
- We **pretend** unit i was treated in periods $T_0 + 1, \dots, T$, impute the “missing” values and compare them to actual values using SC (blue), DID (teal), SDID (red).
- We average squared error by state for 8 periods ($T - T_0 = 8$) to get RMSEs for each state.

TRICKY ASYMPTOTICS

- Under a factor model we need the unit weights to ensure that the α_i for weighted controls is the same as the α_N for treated unit.
- α_i is not observed, and we choose weights to match (weighted) observed lagged outcomes for controls. There are many $(T - 1)$ such lagged outcomes, and only $N - 1$ potential controls. Matching with many matching variables is difficult.
- The formal asymptotic results show that although the matches are not close in terms of the lagged outcomes, they are close in the metric that matters, the unobserved factor α_i . This generalizes to factor models with multiple unobserved factors.

SIMULATIONS

- Common to do simulations with an existing panel dataset, and randomly designate a set of units/time periods to be artificially “treated.”
 - Can then evaluate bias, RMSE of estimates as compared to the observed outcomes.
- Simulations based on complete random assignment can be misleading: don’t catch that estimators may do a bad job picking up the distinct time trends of the treated units.
- Ideal simulations should be “realistic”: outcome model tied to actual data, and assignment process tied to regularities in the assignment of actual treatments.

SIMULATION DESIGN: OUTCOME MODEL

- Follow Bertrand-Duflo-Mullainathan's paper using CPS earnings data.
- For 40 years and 50 states calculate average earnings from CPS. Subtract overall mean, divide by overall standard deviation.
 - Result is matrix $\mathbf{Y}$ with $\|\mathbf{Y}\|_2^2 = 1$.
- Fit rank 4 factor model $\mathbf{M}$. Extract two-way fixed effect matrix from there, $\mathbf{F}$ (based on unit and time fixed effects).
 - Remainder of rank 4 matrix is low rank matrix $\mathbf{L} = \mathbf{M} - \mathbf{F}$.
 - Residuals $\mathbf{E} = \mathbf{Y} - \mathbf{M}$ from rank 4 factor model are used to fit AR(2) model.
- Result from fitting factor model:
 $\|\mathbf{L}\|_2^2 = 0.01, \|\mathbf{F}\|_2^2 = 0.98, \|\mathbf{E}\|_2^2 = 0.01, \rho_1 = 0.01, \rho_2 = -0.06$ (fixed effects explain much of variation, very little autocorrelation left)

SIMULATION DESIGN: TREATMENT ASSIGNMENT PROCESS

- **Goal:** Create a realistic treatment assignment model (no covariates), where the assignment model respects our model assumptions
- Requires treatments independent of unsystematic component of outcomes, but potentially related to time trends or outcome levels for treated units
- To implement:
 - We start with actual regulations (e.g., high state minimum wage, gun law, abortion law)
 - Use logistic regression to predict presence of regulation on four state factors from simulation outcome model (**M**)
 - Goodness of fit shows that treatment assignment responds strongly to unobserved latent factors
 - ★ Minimum wage law $R^2 = 0.30$, gun law $R^2 = 0.15$, abortion law $R^2 = 0.30$.
 - Assign treatment to states with probabilities based on logistic model, with maximum of ten treated states.

RMSE IN SIMULATIONS

	SDID	DID	SC	MC
Main Settings				
Min wage	0.026	0.048	0.040	0.033
$N_1 = 1, T_1 = 10$	0.060	0.127	0.069	0.078
$N_1 = 10, T_1 = 1$	0.055	0.075	0.058	0.055
Gun law	0.023	0.038	0.025	0.028
• Abortion law	0.025	0.044	0.040	0.031
Constrained Assignment				
Random Assignment	0.025	0.042	0.021	0.029
Constrained Outcome Model				
No fixed effects F (favors SC)	0.029	0.045	0.021	0.033
No low rank L (favors DID)	0.017	0.015	0.021	0.015
No noise (no E)	0.002	0.044	0.021	0.003

BIAS IN SIMULATIONS

	SDID	DID	SC	MC
Main Settings				
Min wage	0.007	0.025	0.029	0.018
$N_1 = 1, T_1 = 10$	0.003	0.013	0.027	0.007
$N_1 = 10, T_1 = 1$	0.019	0.039	0.027	0.025
Gun law	0.006	0.007	-0.004	0.009
• Abortion law	0.003	-0.001	0.029	0.000
Constrained Assignment				
Random Assignment	0.001	0.006	0.002	0.003
Constrained Outcome Model				
No fixed effects F (favors SC)	0.006	0.021	0.006	0.013
No low rank L (favors DID)	-0.001	-0.002	0.007	-0.002
No noise (no E)	0.001	0.022	0.006	0.000

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